

# WASP-10b

R-1851

Benchmark run · workflow W8-benchmark-deep · record deep-bench\_WASP-10\_130818 · created 2026-08-03T08:20:57+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

## Results

|                                                   |                                |
|---------------------------------------------------|--------------------------------|
| Mid-Transit Time (Tmid)                           | 2456522.7645 +/- 0.002 BJD_TDB |
| Ratio of Planet to Stellar Radius (Rp/R*)         | 0.211 +/- 0.063                |
| Transit depth (Rp/Rs)^2                           | 4.44 +/- 2.65 [%]              |
| Orbital Inclination (inc)                         | 89.56 +/- 1.89                 |
| Airmass coefficient 1 (a1)                        | 1.52 +/- 1.37                  |
| Airmass coefficient 2 (a2)                        | -0.26 +/- 0.83                 |
| Scatter in the residuals of the lightcurve fit is | 97.37 %                        |
| Best Comparison Star                              | #1 - [131, 267]                |
| Optimal Aperture                                  | 1.34                           |
| Optimal Annulus                                   | 6.25                           |
| Transit Duration (day)                            | 0.094 +/- 0.01                 |

## Accuracy vs published ephemeris (ExoClock/NEA)

|                           |                                           |
|---------------------------|-------------------------------------------|
| Rp/R■ fit vs published    | 0.211 ± 0.063 vs 0.1592 (deviation 0.82σ) |
| Duration fit vs published | 0.094 d vs 0.0946 d (deviation 0.06σ)     |
| Mid-time O–C              | -4.7 min                                  |
| Depth SNR                 | 3.3                                       |
| Residual scatter          | 97.37 %                                   |

## Light curve

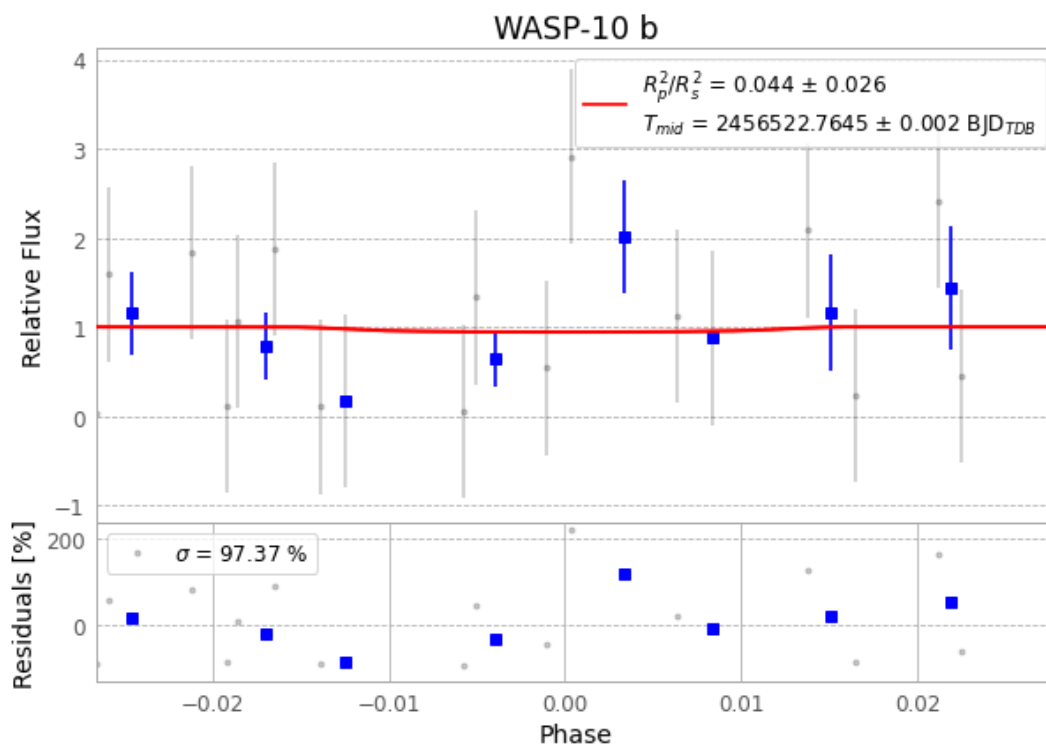


Figure 1 — FinalLightCurve\_WASP-10 b\_2013-08-17.png (SHA-256 aac39da712a481d6...)

## Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [206, 315], 3 comparison star(s). Exact configuration: parameters.inits\_json in record.json (SHA-256 de6b148c7a524c1c...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline\_commit ba3ef8b

## Data provenance

83 FITS frames (55 MB), WASP-10130818041512.FITS → WASP-10130818082112.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-10-130818.log (36dc4313dac5...), FinalLightCurve\_WASP-10 b\_2013-08-17.png (aac39da712a4...), FinalLightCurve\_WASP-10 b\_2013-08-17.csv (cf97a0991843...)

## Compute resources

Wall time 405.7 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-03 08:20Z.